COMPOSITIONS OF ETALE MORPHISMS ARE ETALE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Etaleness is a local property of ring maps. Lemma \ref{lemma-etale-characterize}, cited at proof lines 8126-8128. Reduces to ring maps.
- Permanence of a property of ring maps under composition, and stability of etale ring maps under composition. Lemma \ref{lemma-composition-type-P} and Algebra, Lemma \ref{algebra-lemma-etale}, cited at proof lines 8128-8133. Give the composition statement.

Source reference. Original lemma 02GN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3234–3234. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 02GJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3230–3230. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01SV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2352–2352. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00U2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1096–1096. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *The composition of two morphisms which are étale is étale.*

Proof. In the proof of Lemma 02GJ we saw that being étale is a local property of ring maps. Hence the first statement of the lemma follows from Lemma 01SV combined with the fact that being étale is a property of ring maps that is stable under composition, see Algebra, Lemma 00U2. □